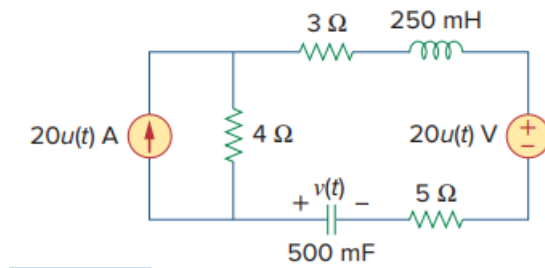


Problem

Determine $v(t)$ for $t > 0$ in the circuit of Fig. 8.87.

Figure 8.87:



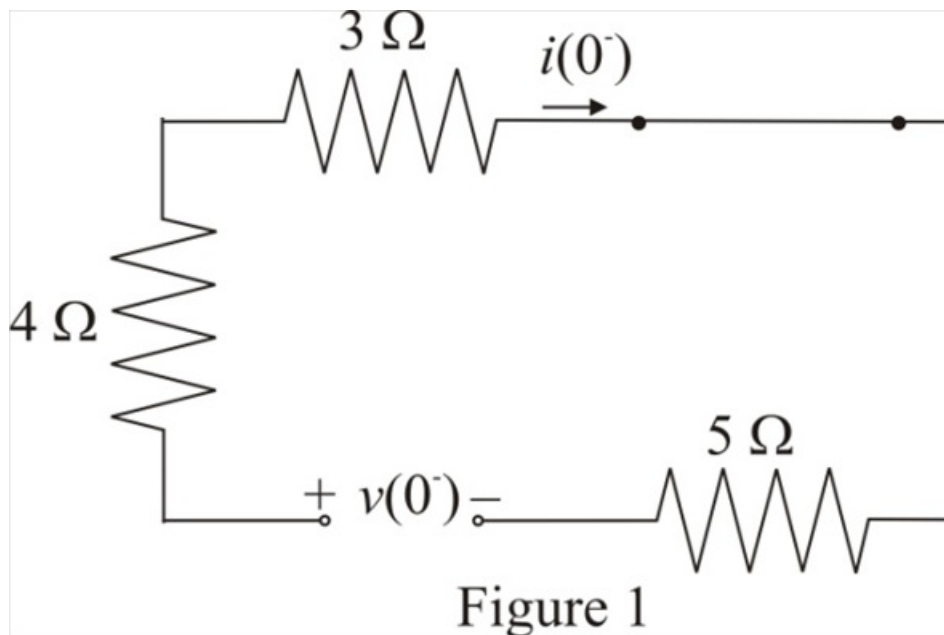
Step-by-step solution

Step 1 of 14

Refer to Figure 8.87 in the text book.

At $t < 0$, the capacitor acts as open circuit and inductor acts as short circuit. The value of the current source is 0 A and the value of the voltage source is 0 V since $u(t)$ is 0 at $t < 0$.

Draw the circuit diagram as shown in Figure 1.

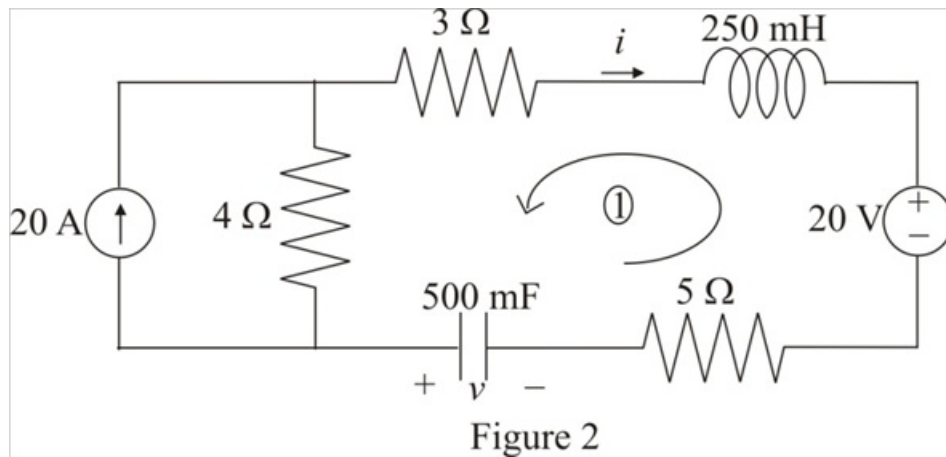


The values of the current $i(0^-)$ and the voltage $v(0^-)$ is 0 A and 0 V respectively.

Step 2 of 14

Draw the circuit diagram at $t > 0$.

The value of the current source is 20 A and the value of the voltage source is 20 V since $u(t)$ is 1 at $t > 0$.



Step 3 of 14

The voltage across the capacitor and the current through the inductor will not change instantaneously. Hence, at $t > 0$ the value of the voltage across the capacitor $v(0^+)$ is 0 V and the value of the current $i(0^+)$ is 0 A .

Notice that the voltage across the capacitor and the current through the inductor are continuous. Therefore, the values at $t = 0$ are $v(0) = 0\text{ V}$ and $i(0) = 0\text{ A}$.

Step 4 of 14

Consider the expression for the current passing through the capacitor.

$$i = (500\text{ m}) \frac{dv}{dt}$$

$$\frac{dv}{dt} = 2i$$

Calculate the value of $\frac{dv}{dt}$ for $t = 0$.

$$\frac{dv(0)}{dt} = 2i(0)$$

Substitute 0 A for $i(0)$.

$$\begin{aligned} \frac{dv(0)}{dt} &= 2(0) \\ &= 0\text{ V/s} \end{aligned}$$

Therefore, the value of $\frac{dv(0)}{dt}$ is 0 V/s .

Step 5 of 14

Apply Kirchhoff's Voltage Law in loop-1.

$$-20 + (250 \text{ m}) \frac{di}{dt} + 3i + 4(i + 20) + \frac{1}{500 \text{ m}} \int i \, dt + 5i = 0$$

$$-20 + (250 \text{ m}) \frac{di}{dt} + 3i + 4i + 80 + 2 \int i \, dt + 5i = 0$$

$$(250 \text{ m}) \frac{di}{dt} + 12i + 60 + 2 \int i \, dt = 0$$

Substitute $(500 \text{ m}) \frac{dv}{dt}$ for i .

$$(250 \text{ m}) \frac{d}{dt} \left((500 \text{ m}) \frac{dv}{dt} \right) + 12 \left((500 \text{ m}) \frac{dv}{dt} \right) + 60 + 2 \int \left((500 \text{ m}) \frac{dv}{dt} \right) dt = 0$$

$$(125 \text{ m}) \frac{d^2 v}{dt^2} + 6 \frac{dv}{dt} + 60 + \int dv = 0$$

$$(125 \text{ m}) \frac{d^2 v}{dt^2} + 6 \frac{dv}{dt} + 60 + v = 0$$

$$\frac{d^2 v}{dt^2} + 48 \frac{dv}{dt} + 480 + 8v = 0$$

$$\frac{d^2 v}{dt^2} + 48 \frac{dv}{dt} + 8v = -480 \dots\dots (1)$$

Step 6 of 14

To estimate the transient analysis of the circuit, consider equation (1) to be homogeneous.

$$\frac{d^2 v}{dt^2} + 48 \frac{dv}{dt} + 8v = 0$$

Consider $\frac{d}{dt}$ to be s .

$$s^2 v + 48sv + 8v = 0$$

$$(s^2 + 48s + 8)v = 0$$

Step 7 of 14

Consider the characteristic equation of the circuit.

$$s^2 + 48s + 8 = 0$$

$$(s + 0.1673)(s + 47.8327) = 0$$

Therefore, the roots of the characteristic equation are $s_1 = -0.1673$ and $s_2 = 47.8327$.

The roots of the characteristic equation are negative and real. Hence, the circuit exhibits over-damped response.

Step 8 of 14

Draw the circuit diagram for $t = \infty$.

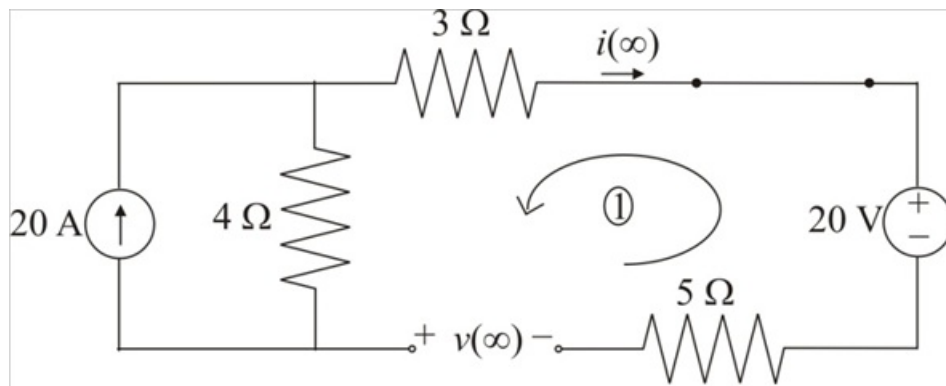


Figure 3

Step 9 of 14

The value of the current $i(\infty)$ is 0 A. For $i(\infty)$.

Apply Kirchhoff's Voltage Law in loop-1.

$$-20 + 3i(\infty) + 4(i(\infty) + 20) + v(\infty) + 5i(\infty) = 0$$

$$-20 + 3i(\infty) + 4i(\infty) + 80 + v(\infty) + 5i(\infty) = 0$$

$$12i(\infty) + 60 + v(\infty) = 0$$

$$v(\infty) = -60 - 12i(\infty)$$

Substitute 0 A for $i(\infty)$.

$$v(\infty) = -60 - 12(0)$$

$$= -60 - 0$$

$$= -60 \text{ V}$$

Therefore, the value of the voltage $v(\infty)$ is -60 V.

Step 10 of 14

Consider the expression or the voltage $v(t)$.

$$v(t) = v(\infty) + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Substitute -60 V for $v(\infty)$, -47.8327 for s_1 and -0.1673 for s_2 .

$$v(t) = -60 + A_1 e^{-47.8327t} + A_2 e^{-0.1673t} \dots\dots (2)$$

Step 11 of 14

Recall equation (2).

$$v(t) = -60 + A_1 e^{-47.8327t} + A_2 e^{-0.1673t}$$

Substitute 0 for t .

$$\begin{aligned} v(0) &= -60 + A_1 e^{-47.8327(0)} + A_2 e^{-0.1673(0)} \\ &= -60 + A_1 e^0 + A_2 e^0 \\ &= -60 + A_1 (1) + A_2 (1) \\ &= -60 + A_1 + A_2 \end{aligned}$$

Substitute 0 V for $v(0)$.

$$\begin{aligned} 0 &= -60 + A_1 + A_2 \\ A_2 &= 60 - A_1 \dots\dots (3) \end{aligned}$$

Step 12 of 14

Recall equation (1).

$$v(t) = -60 + A_1 e^{-47.8327t} + A_2 e^{-0.1673t}$$

Apply derivative with respect to t .

$$\begin{aligned} \frac{dv(t)}{dt} &= \frac{d}{dt}(-60) + A_1 \frac{d}{dt}(e^{-47.8327t}) + A_2 \frac{d}{dt}(e^{-0.1673t}) \\ &= 0 - 47.8327 A_1 e^{-47.8327t} - 0.1673 A_2 e^{-0.1673t} \\ &= -47.8327 A_1 e^{-47.8327t} - 0.1673 A_2 e^{-0.1673t} \end{aligned}$$

Substitute 0 for t and $60 - A_1$ for A_2 .

$$\begin{aligned} \frac{dv(0)}{dt} &= -47.8327 A_1 e^{-47.8327t} - 0.1673(60 - A_1) e^{-0.1673t} \\ &= -47.8327 A_1 e^0 - 0.1673(60 - A_1) e^0 \\ &= -47.8327 A_1 (1) - (10.038 - 0.1673 A_1)(1) \\ &= -47.8327 A_1 - 10.038 + 0.1673 A_1 \end{aligned}$$

Substitute 0 V/s for $\frac{dv(0)}{dt}$

$$\begin{aligned} 0 &= -47.6654 A_1 - 10.038 \\ 47.6654 A_1 &= -10.038 \\ A_1 &= -\frac{10.038}{47.6654} \\ A_1 &= -0.21 \end{aligned}$$

Therefore, the value of the constant A_1 is -0.21 .

Step 13 of 14

Recall equation (3).

$$A_2 = 60 - A_1$$

Substitute -0.21 for A_1 .

$$\begin{aligned} A_2 &= 60 - (-0.21) \\ &= 60.21 \end{aligned}$$

Therefore, the value of the constant A_2 is 60.21 .

Step 14 of 14

Recall equation (1).

$$v(t) = -60 + A_1 e^{-47.8327t} + A_2 e^{-0.1673t}$$

Substitute 0.21 for A_1 and 60.21 for A_2 .

$$v(t) = -60 - 0.21e^{-47.8327t} + 60.21e^{-0.1673t} \text{ V}$$

Therefore, the value of the voltage $v(t)$ is $\boxed{-60 - 0.21e^{-47.8327t} + 60.21e^{-0.1673t} \text{ V}}$.

[Comments \(1\)](#)



Anonymous

The solution changed $s_1 = -.1673$ to $s_1 = -47.8327$. This changes the final answer entirely. But, doing so gives the same answer as in the back of the book. Perhaps, both solution answers are incorrect.
